FormaProof - verified publication excerpt

US20080003142A1 - Microfluidic devices

Original publication: 3 January 2008.

This file contains original PDF pages 1, 2 and 81 of 83.

It is an excerpt, not the complete patent publication.

The original has been retrieved from the PDF linked by the user-provided Google Patents page.

Source: https://patents.google.com/patent/US20080003142A1/en

PDF: https://patentimages.storage.googleapis.com/89/d6/88/68356f0ae820d5/US20080003142A1.pdf

Original SHA-256: 7f2fdf36b240bc60d0fb6745369acdbe11198910595f9690037b69214dca661c

Source checked: 2026-10-25. Legal status is not independently verified.

This example is a known-reference comparison, not a fresh worldwide search.

Published technical content does not itself establish active patent rights or permission to manufacture.

US 20080003142A1

(19) **United States**

(12) **Patent Application Publication** (10) **Pub. No.: US 2008/0003142 A1**

Link et al. (43) **Pub. Date: Jan. 3, 2008**

(54) **MICROFLUIDIC DEVICES**

(76) Inventors: **Darren R. Link**, Guilford, CT (US); **Michael Weiner**, Guilford, CT (US); **David Marran**, Durham, CT (US); **Jonathan M. Rothberg**, Guilford, CT (US)

Correspondence Address:
MINTZ, LEVIN, COHN, FERRIS, GLOVSKY AND POPEO, P.C.
ONE FINANCIAL CENTER
BOSTON, MA 02111 (US)

(21) Appl. No.: **11/803,104**

(22) Filed: **May 11, 2007**

Related U.S. Application Data

(60) Provisional application No. 60/799,833, filed on May 11, 2006. Provisional application No. 60/799,834, filed on May 11, 2006. Provisional application No. 60/808,614, filed on May 25, 2006. Provisional application No. 60/815,097, filed on Jun. 19, 2006. Provisional application No. 60/819,733, filed on Jul. 7, 2006. Provisional application No. 60/819,734, filed on Jul. 7, 2006. Provisional application No. 60/841,716, filed on Sep. 1, 2006. Provisional application No. 60/843,374, filed on Sep. 8, 2006. Provisional application No. 60/833,151, filed on Jul. 24, 2006. Provisional application No. 60/834,987, filed on Jul. 31, 2006. Provisional application No. 60/837,871, filed on Aug. 14, 2006. Provisional application No. 60/837,695, filed on Aug. 14, 2006. Provisional application No. 60/843,327, filed on Sep. 11, 2006. Provisional application No. 60/856,540, filed on Nov. 3, 2006. Provisional application No. 60/856,440, filed on Nov. 3, 2006. Provisional application No. 60/874,561, filed on Dec. 12, 2006. Provisional application No. 60/858,279, filed on Nov. 8, 2006. Provisional application No. 60/858,278, filed on Nov. 8, 2006. Provisional application No. 60/874,640, filed on Dec. 12, 2006. Provisional application No. 60/860,665, filed on Nov. 22, 2006. Provisional application No. 60/873,766, filed on Dec. 8, 2006. Provisional application No. 60/876,209, filed on Dec. 20, 2006. Provisional application No. 60/899,258, filed on Feb. 2, 2007. Provisional application No. 60/903,153, filed on Feb. 23, 2007. Provisional application No. 60/904,293, filed on Feb. 28, 2007. Provisional application No. 60/920,337, filed on Mar. 26, 2007.

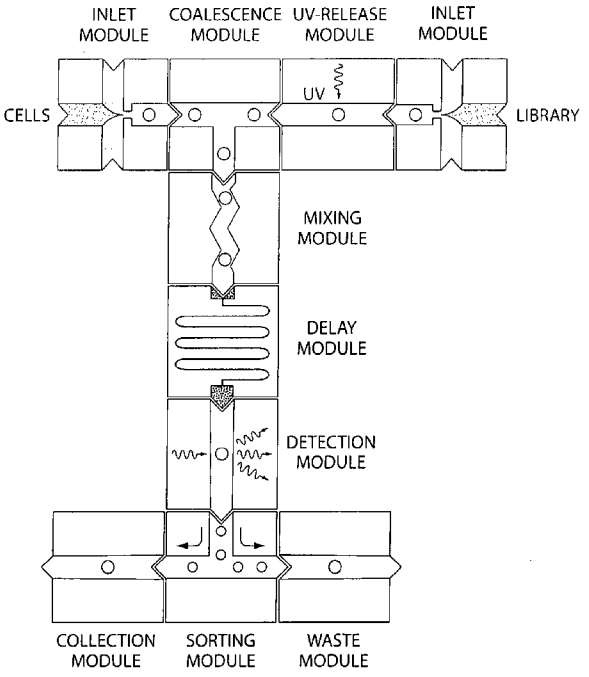
Publication Classification

(51) **Int. Cl.**
G01N 21/64 (2006.01)
B01L 3/00 (2006.01)
B29C 33/38 (2006.01)

(52) **U.S. Cl.** **422/82.08**; 264/219; 422/100

(57) **ABSTRACT**

The present invention provides novel microfluidic substrates and methods that are useful for performing biological, chemical and diagnostic assays. The substrates can include a plurality of electrically addressable, channel bearing fluidic modules integrally arranged such that a continuous channel is provided for flow of immiscible fluids.



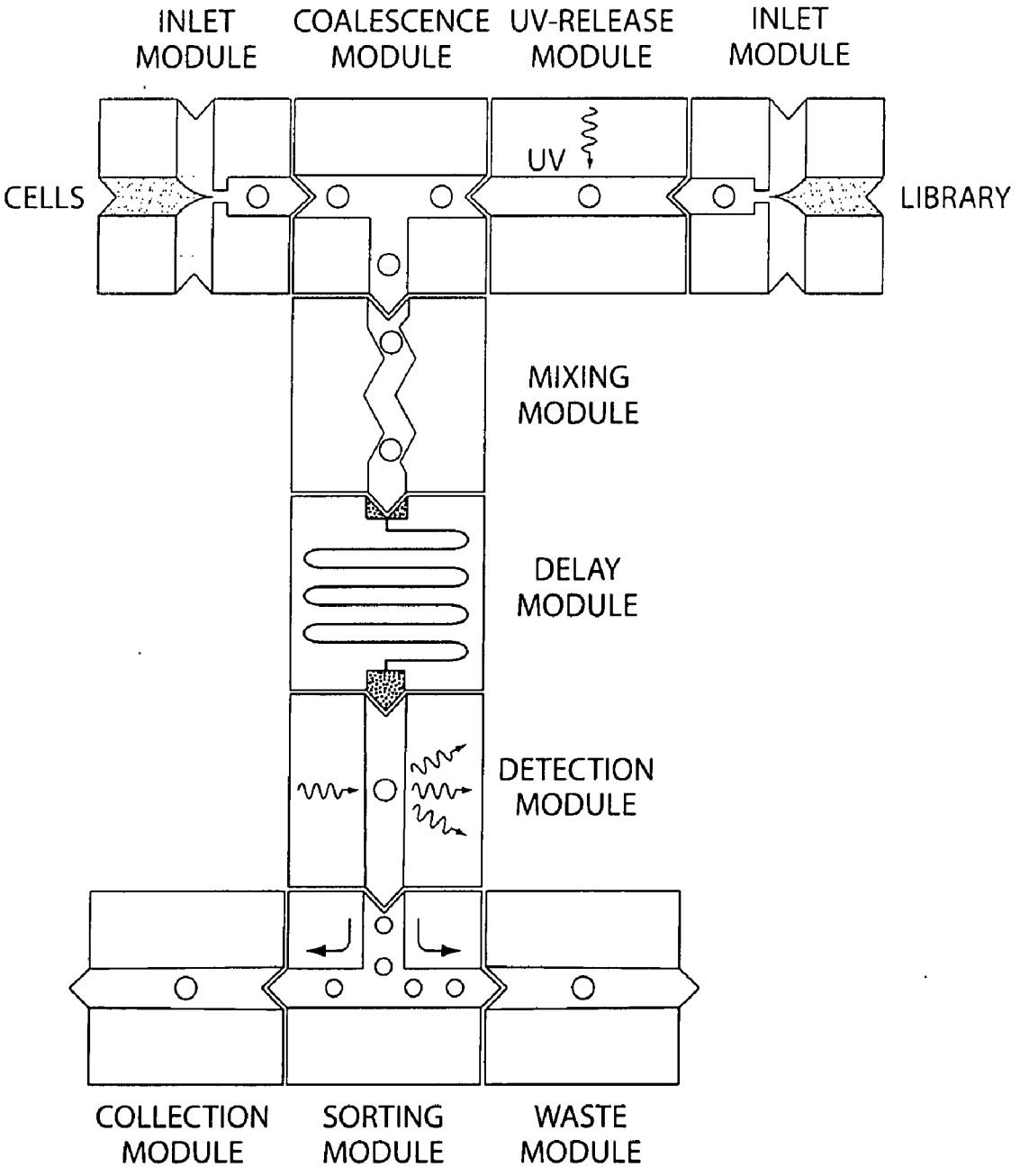

INLET MODULE
COALESCENCE MODULE
UV-RELEASE MODULE
INLET MODULE
UV
CELLS
LIBRARY
MIXING MODULE
DELAY MODULE
DETECTION MODULE
COLLECTION MODULE
SORTING MODULE
WASTE MODULE

Fig. 1

[0439] In other embodiments, the substrate can be added with—or simultaneously formulated from two or more separate reagent streams with the contained within a droplet. For a specific example, macrophage cells are washed in buffer and incubated with an enzyme-conjugated anti-cell-surface antibody. The cells are then individually loaded into droplets formulated at the same time, with the enzyme substrate. The amount of enzyme substrate turned over within the droplet will be proportional to the number of enzyme molecules within the droplet, which is proportional to the number of antibodies bound to the macrophage surface. By careful calibration it should be possible to estimate the number of cell-surface molecules attached to the cell surface.

What is claimed is:

1. A microfluidic substrate comprising:

a) at least one inlet module having at least one inlet channel adapted to carry at least one dispersed phase fluid; and

b) at least one main channel adapted to carry at least one continuous phase fluid, wherein said inlet channel is in fluid communication with said main channel at a junction, wherein said junction comprises a fluidic nozzle designed for flow focusing such that said dispersed phase fluid is immiscible with said continuous phase fluid and forms a plurality of highly uniform, monodisperse droplets in said continuous phase fluid.

2. The microfluidic substrate of claim 1, wherein said inlet module further comprises at least one self-aligning fluidic interconnect apparatus to connect the inlet channel to a means for introducing a sample fluid to said channel, wherein said apparatus forms a radial seal between said microfluidic substrate and said means for introducing sample.

3. The microfluidic substrate of claim 2, wherein said means is a well or reservoir.

4. The microfluidic substrate of claim 1, further comprising at least one coalescence module downstream from and in fluid communication with said inlet module via the main channel comprising a coalescence apparatus, wherein two or more droplets passing there through are coalesced to form a nanoreactor.

5. The microfluidic substrate of claim 4, wherein the coalescence apparatus comprises one or more electrodes that generate an electric field.

6. The microfluidic substrate of claim 5, wherein the electrodes comprise electrically conductive material that is integrally contained in one or more channels and isolated from the inlet and main channels.

7. The microfluidic substrate of claim 4, wherein said coalescence module comprises an expanded portion of the main channel between the electrodes to bring successive droplets into proximity, whereby the paired droplets are coalesced within the electric field.

8. The microfluidic substrate of claim 4, wherein said coalescence module comprises a narrowed portion of the main channel to center droplets within the main channel followed by a expanded portion of the main channel between the electrodes to bring successive droplets into proximity, whereby the paired droplets are coalesced within the electric field.

9. The microfluidic substrate of claim 1, wherein said substrate further comprises at least one detection module downstream from and in fluid communication with said coalescence module, said detection module comprising a detection apparatus for evaluating the contents or characteristics of the droplet or nanoreactor.

10. The microfluidic substrate of claim 9, wherein said detection apparatus comprises an optical or electrical detector.

11. The microfluidic substrate of claim 1, wherein said substrate further comprises a sorting module in proximity to and in fluid communication with said detection module, said sorting module comprising a sorting apparatus adapted to direct said droplet or nanoreactor into or away from a collection module in response to the contents or characterization of the droplet or nanoreactor evaluated in the detection module.

12. The microfluidic substrate of claim 11, wherein the sorting apparatus comprises one or more electrodes that generate an electric field.

13. The microfluidic substrate of claim 12, wherein the electrodes comprise an electrically conductive material that is integrally contained in one or more channels and are isolated from the inlet and main channels.

14. The microfluidic substrate of claim 1, wherein said inlet and main channels are coated with an anti-wetting or blocking agent for the dispersed phase.

15. The microfluidic substrate of claim 1, wherein said channels are coated with a silica primer layer followed by a perfluoroalkylalkylsilane compound, an amorphous soluble perfluoropolymer, BSA, PEG-silane or fluorosilane.

16. The microfluidic substrate of claim 1, wherein the channels are coated with a silica primer layer followed by a perfluoroalkylalkylsilane compound.

17. The microfluidic substrate of claim 11, wherein the channels in the sorting module comprise an asymmetric bifurcation geometry.

18. The microfluidic substrate of claim 11, wherein the channels in the sorting module comprise an asymmetric flow geometry.

19. The microfluidic substrate of claim 1, wherein the channels comprise well-like indentations to slow, stop or react contents of droplets.

20. The microfluidic substrate of claim 1, wherein said substrate further comprises a delay module in fluid communication with the main channel downstream of the inlet module.

21. The microfluidic substrate of claim 20, wherein said delay module is a delay line.

22. The microfluidic substrate of claim 20, wherein the delay module further comprises heating and cooling regions.

23. The microfluidic substrate of claim 1, wherein said substrate further comprises a mixing module in fluid communication with the main channel downstream of the inlet module.

24. The microfluidic substrate of claim 1, wherein said substrate further comprises a UV-release module in fluid communication with the main channel downstream of the inlet module.

25. The microfluidic substrate of claim 1, wherein said substrate further comprises a collection module connected to a means for storing a sample from said substrate.

26. The microfluidic substrate of claim 25, wherein said means is a well or reservoir.

27. The microfluidic substrate of claim 1, wherein said substrate further comprises a waste module connected to a means for collecting a sample discarded from said substrate.